

Unveiling the Intrinsic SN Ia Mass Step with BayeSN

Matt Grayling

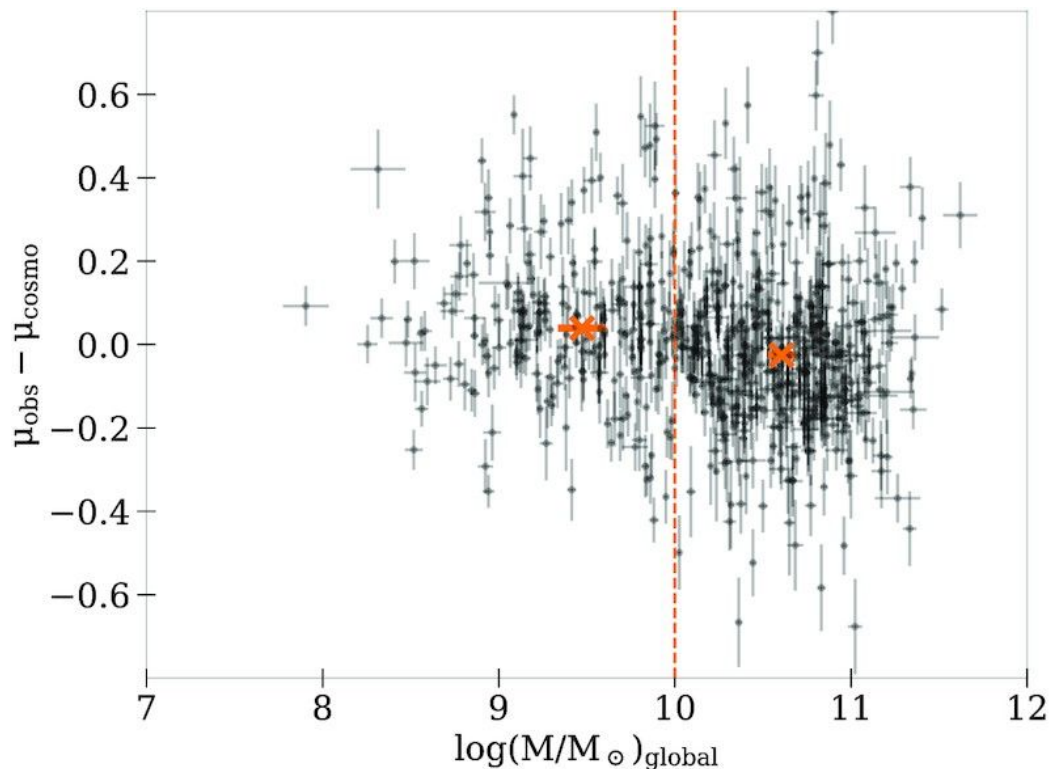


$$\mu^s = m_B^s - M + \alpha x_1^s + \beta c^s$$

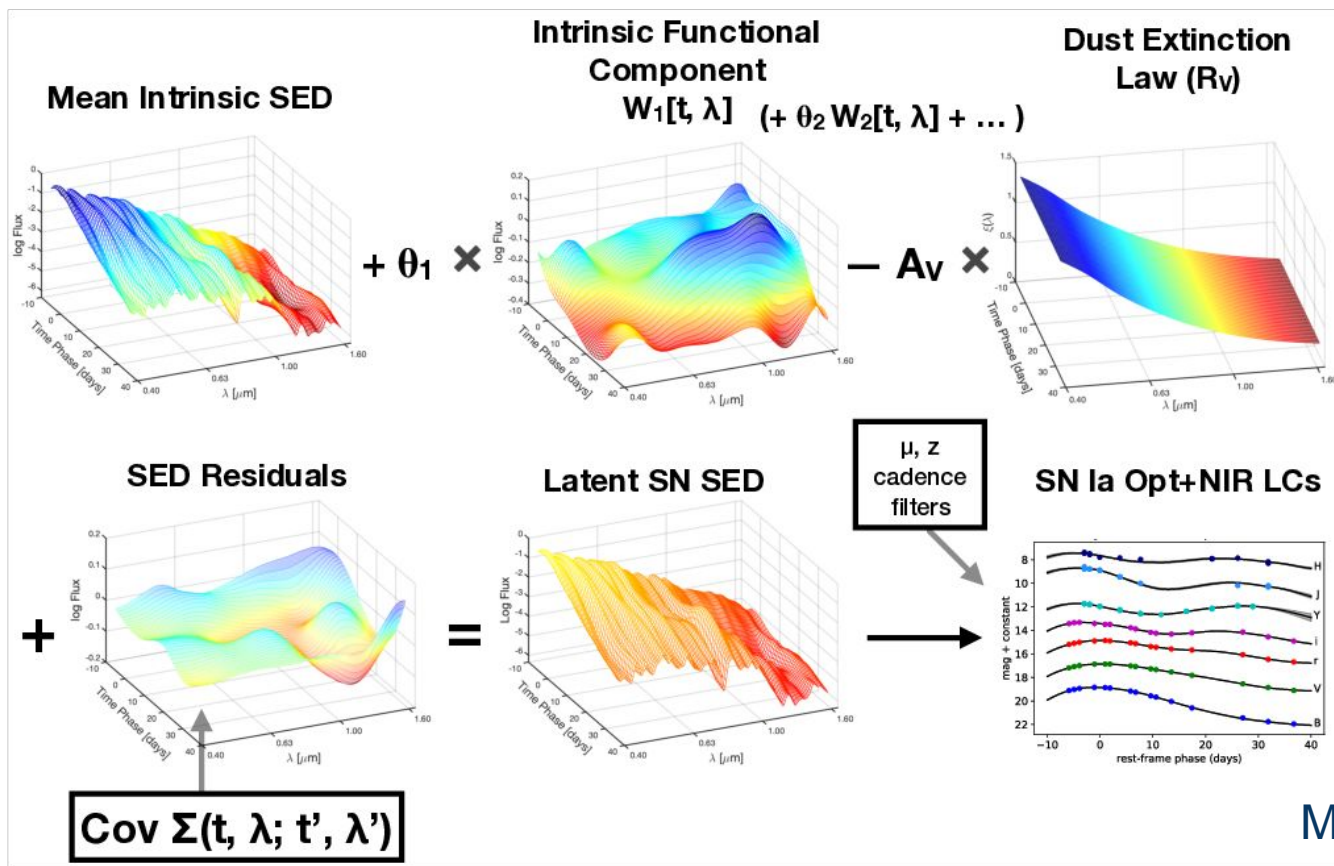
- Tripp formula - one parameter for two separate effects
- Correctly handling dust is key for both SN Ia astrophysics and cosmology
- Dust was the largest source of systematic uncertainty in the DES5YR cosmology result

The 'Mass Step'

- Disentangling intrinsic variation from dust vital to understanding host galaxy 'mass step'
- Just different dust properties?
 - SN populations identical, just appear different?
- Or intrinsic differences?
 - Interesting physical implications, different progenitor channels?

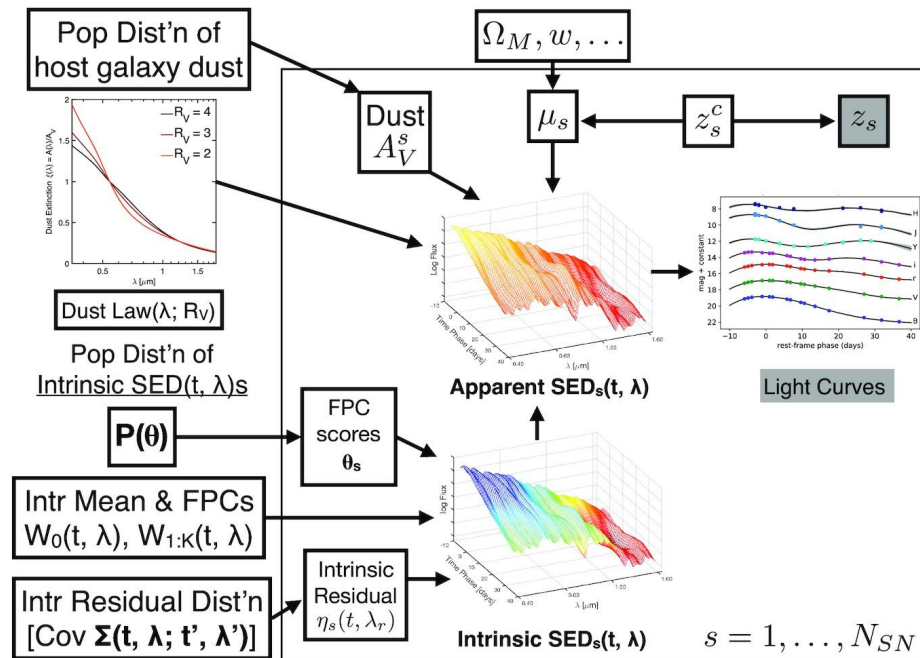


The BayeSN Model



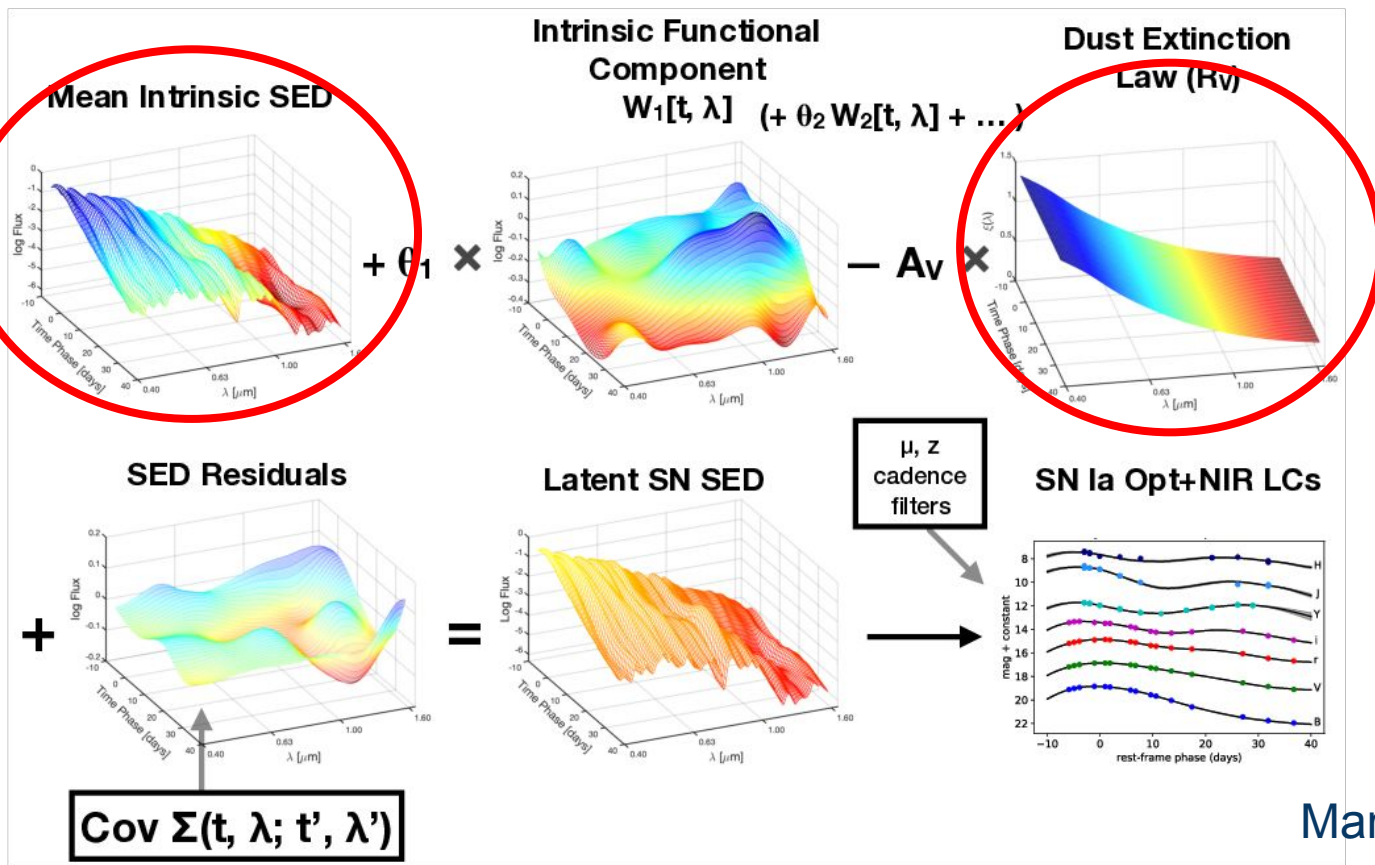
Advantages of BayeSN

- Hierarchical Bayesian Model
 - Allows for separate treatment of dust and intrinsic variation
- Full SED model
 - Uses full light curve in all bands rather than just single colour/magnitude pairs
- Fit for distance jointly with stretch/dust
 - No Tripp formula, distance marginalised over dust/intrinsic variation



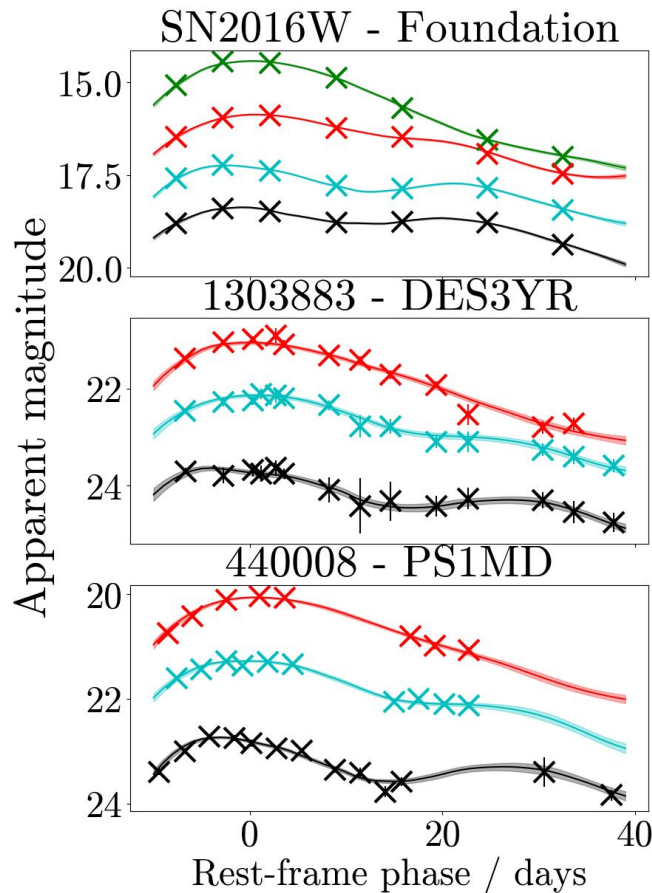
Mandel+2020

Investigating the Mass Step



Investigating the Mass Step

- Infer properties of SN populations separately on each side of the mass step using BayeSN
- Combined sample of 475 SNe Ia from Foundation, DES3YR and PS1MD
- Jointly infer dust and intrinsic differences, see which is more consistent with data

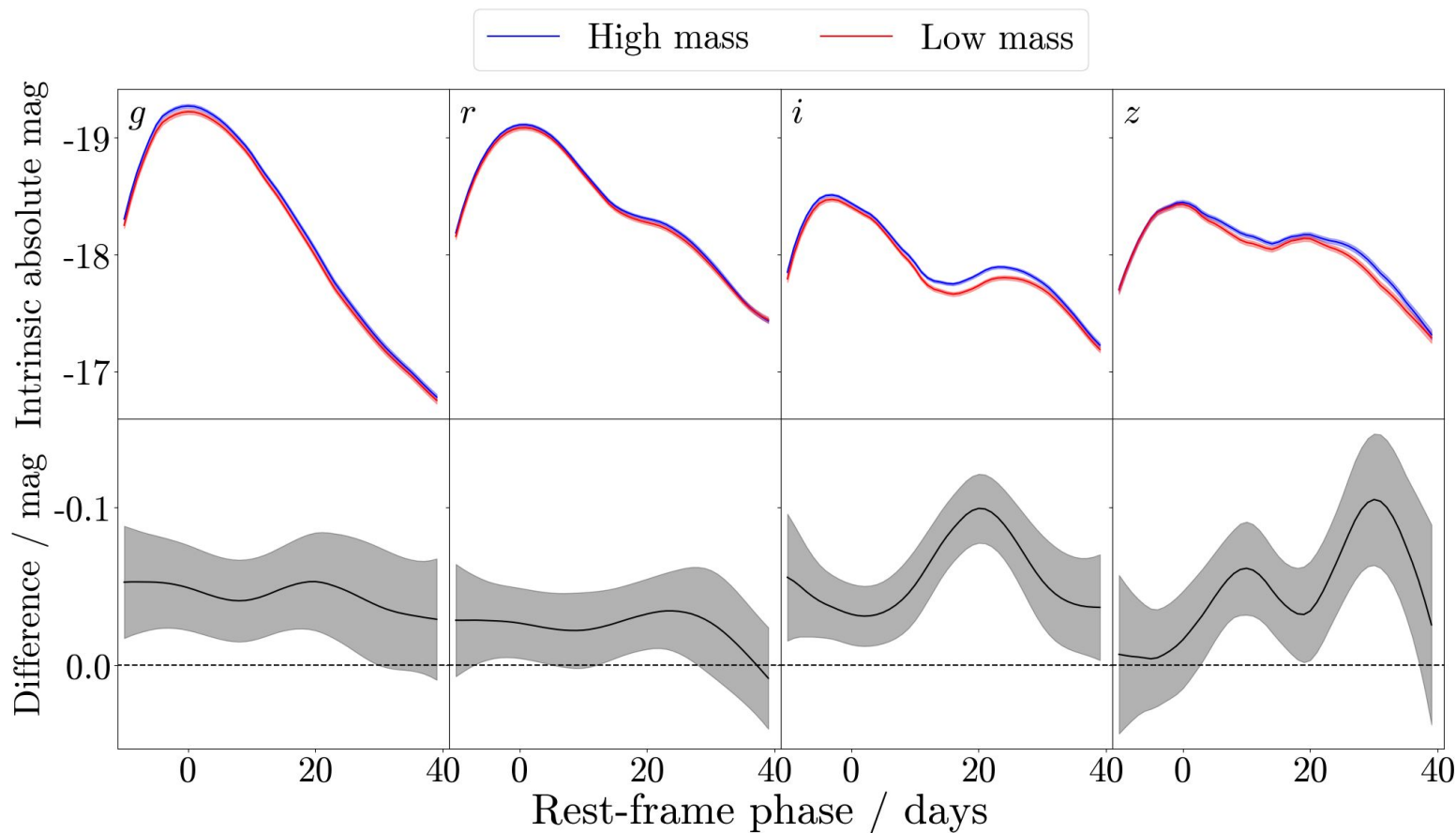


Example
BayeSN
fits

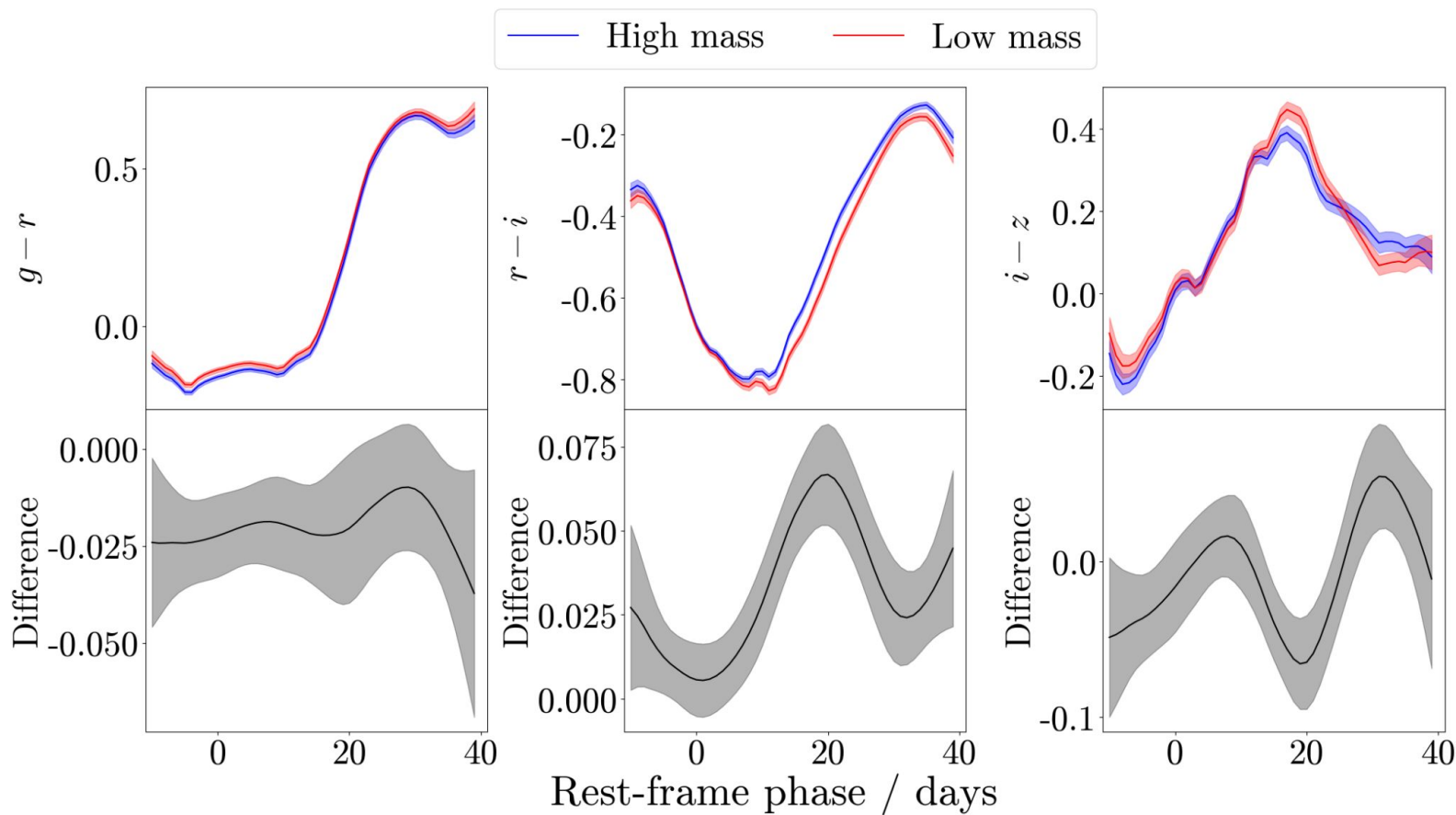
Method	ΔM_0 / mag	$\Delta(g - r)_{\text{peak}}^{\text{int}}$	μ_R		τ_A / mag	
			HM	LM	HM	LM
Intrinsic mag diff.	-0.049 ± 0.016	—	2.51 ± 0.16	2.74 ± 0.35	0.18 ± 0.01	0.11 ± 0.01
Intrinsic SED diff.	-0.049 ± 0.027	-0.022 ± 0.010	2.26 ± 0.14	3.36 ± 0.51	0.19 ± 0.02	0.15 ± 0.02

- Our results support the existence of intrinsic differences between SNe Ia in each mass bin
- Depending on our assumptions, our results also indicate differences in host galaxy dust properties

Intrinsic Baseline Light Curves



Intrinsic Baseline Colour Curves



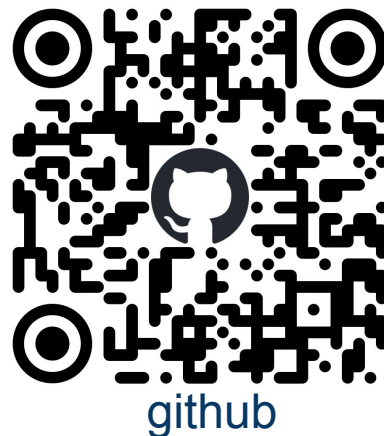
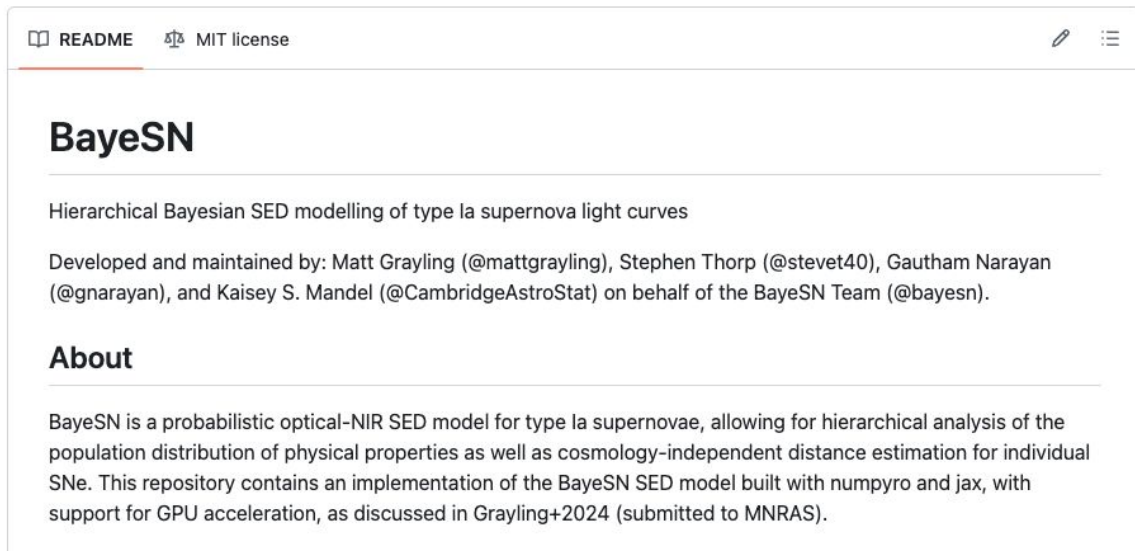
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- Our results support the existence of intrinsic differences between SNe Ia in each mass bin
- Depending on our assumptions, our results also indicate differences in host galaxy dust properties
- Lots of different, SED-level effects. Analyses of larger samples will be able to further constrain the relative balance of intrinsic and extrinsic effects

BayeSN Out Now!

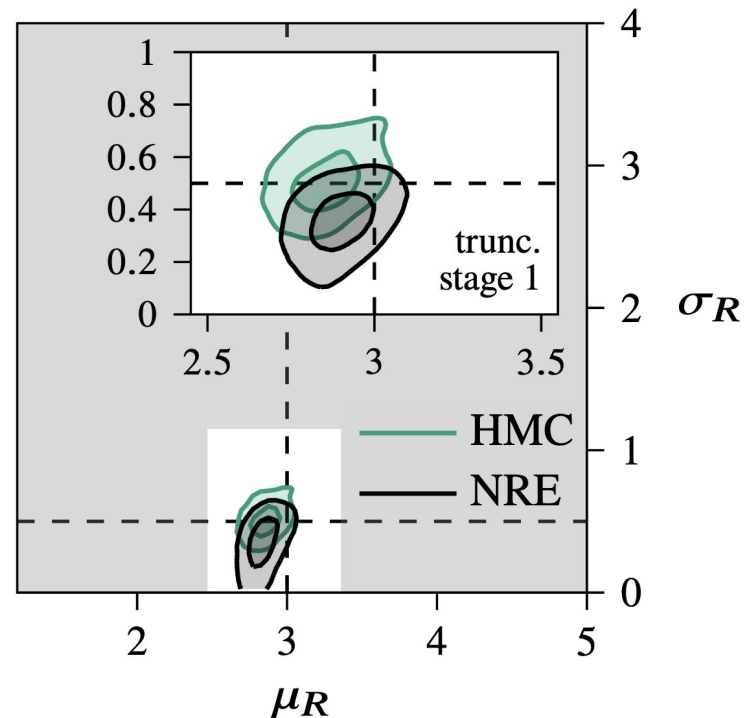
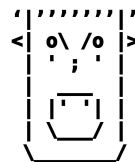
New GPU-accelerated, Python implementation
of BayeSN out now!

github.com/bayesn/bayesn, pip installable,
SNANA integration in progress



The Future of BayeSN

- Accelerating BayeSN light curve fits using variational inference (see Uzsoy+2024, arXiv:2405.06013)
- SNANA integration in progress
- Incorporating spectra in training is planned!
- Fully hierarchical cosmology?!
 - Simulation-based inference (SBI) may provide the solution!
 - See Karchev, Grayling et al. (2024) for an application to SN dust inference



- Using BayeSN, we can use the full time and wavelength evolution of SN Ia light curves to separately constrain intrinsic and extrinsic differences between mass bins
- We find evidence in favour of intrinsic differences, supporting the idea of different SN Ia populations in different environments
- Extrinsic differences may also play a role. Future analyses with larger samples will be able to better constrain the relative size of these effects
- BayeSN out now, pip installable, can also be used to infer distances for cosmological analyses